TEAVISION360 – A COMPLETE AI AND IOT PLATFORM FOR QUALITY, HEALTH AND MARKET DECISIONS IN TEA FARMING

Project ID: RP [25-26J-178]

Project Proposal Report

Weerawardane K W R C - IT22239402

Malshan W A D D I -IT22137050

Jayasundara K B S M - IT22241832

Udawela U K R M R N - IT22271396

B.Sc. (Hons) Degree in Information Technology Specialization in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

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TEA MARKET ANALYSIS AND PRICE FORECASTING WITH A SMART DASHBOARD FOR DECISION SUPPORT.

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Jayasundara K B S M – IT22241832

Supervisor – Ms. Dushanthi Kuruppu

Co-Supervisor – Ms. Chathurya Kumarapperuma

B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology
Sri Lanka

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Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Group Member Name	Student ID	Signature
K B S M Jayasundara	IT22241832	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Co Supervisor:
(Ms. Chathurya Kumarapperuma)

Date: 29/08/2025

Abstract

The tea industry is a backbone of the Sri Lankan economy, driving foreign exchange earnings, employment, and global recognition as a producer of high-quality tea. The industry, however, faces intricate issues like unstable international market conditions, climate-induced production variability, increasing competition from new producers, and changing consumer preferences. While the big exporters attempt to address these challenges with limited data-backed strategies, the small and medium-scale producers who make up the majority of the sector remain underserved by the lack of readily available, reliable market information. The dearth of information constrains informed decision-making, risk management, and strategic planning along the value chain.

This research bridges this critical gap by developing an AI-powered decision support system for Sri Lanka's tea market. The system integrates disparate internal and external datasets production volumes, export statistics, global price indices, climatic patterns, and consumer demand trends into a unified analytical framework. Advanced regression and time-series predictive models are employed to discern trends, predict price movements, and evaluate the effects of exogenous shocks such as climatic fluctuations, trade policy reversals, and global market disruptions.

The system outputs are displayed in an intelligent, web-based dashboard, customized for all stakeholders, from smallholder farmers and local traders to large exporters and policymakers. With real-time price forecasting, trend analysis, and actionable strategic suggestions, the platform enables informed decision-making, reduces financial risks, and enhances market competitiveness.

The anticipated outcome of this project is the democratization of access to high-end analytics, transparency in the tea supply chain, and driving digital transformation in one of Sri Lanka's most significant economic sectors. By combining AI innovation with stakeholder-centered design, the platform not only drives resilience and efficiency but also positions Sri Lanka at the cutting edge of leveraging advanced technology for agricultural markets.

Keywords: Tea market analysis, Price forecasting, Artificial Intelligence, Intelligent dashboard, Time-series forecasting, Decision support system, Digitalization

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List of Abbreviations

- **FAO** – Food and Agriculture Organization
- **SECM** – Seasonal Error Correction Model
- **VEC** – Vector Error Correction
- **ARIMA** – AutoRegressive Integrated Moving Average
- **SARIMA** – Seasonal AutoRegressive Integrated Moving Average
- **ANN** – Artificial Neural Network
- **ML** – Machine Learning
- **AI** – Artificial Intelligence
- **MLP** – Multi-Layer Perceptron
- **XAI** – Explainable Artificial Intelligence
- **MDPI** – Multidisciplinary Digital Publishing Institute
- **VaR** – Value at Risk
- **IISD** – International Institute for Sustainable Development
- **SHAP** – SHapley Additive exPlanations

1. Introduction

1.1 Background & Literature Review

Sri Lanka's tea industry marketed internationally as "Ceylon Tea" has supported the country's economy for over a century, making a significant foreign exchange and rural livelihood contribution. Current official statistics emphasize its macroeconomic impact and stresses today. The Sri Lanka Tea Board's 2023 Annual Report notes Sri Lanka producing approximately 5% of global tea, but with a disproportionately high share in exports; however, industry margins are narrowing due to softness in demand, input cost inflation, and aggressive competition from India and Kenya. The report further mentions persistent exposure to exchange-rate volatility and fertilizer and fuel cost shocks. Sri Lanka Tea Board.

At the same time, global monitoring by the FAO Intergovernmental Group on Tea observed that after a 14.5% hike in 2022, average auction prices fell by 9.2% in 2023 in the presence of increased availabilities and slowed demand growth terms that enhance volatility for production countries like Sri Lanka.

Sri Lanka's price discovery is led by the Colombo Tea Auction, known to be the oldest serially running tea auction (fitted up for auction for the first time on 30 July 1883). Its strength is transparency and market reach; its weakness, from a planning perspective, is that it is not predictive. Stakeholders especially smallholders and local traders lack a planning deficit due to the fact that the auction is a contemporaneous mechanism and not a forward mechanism.

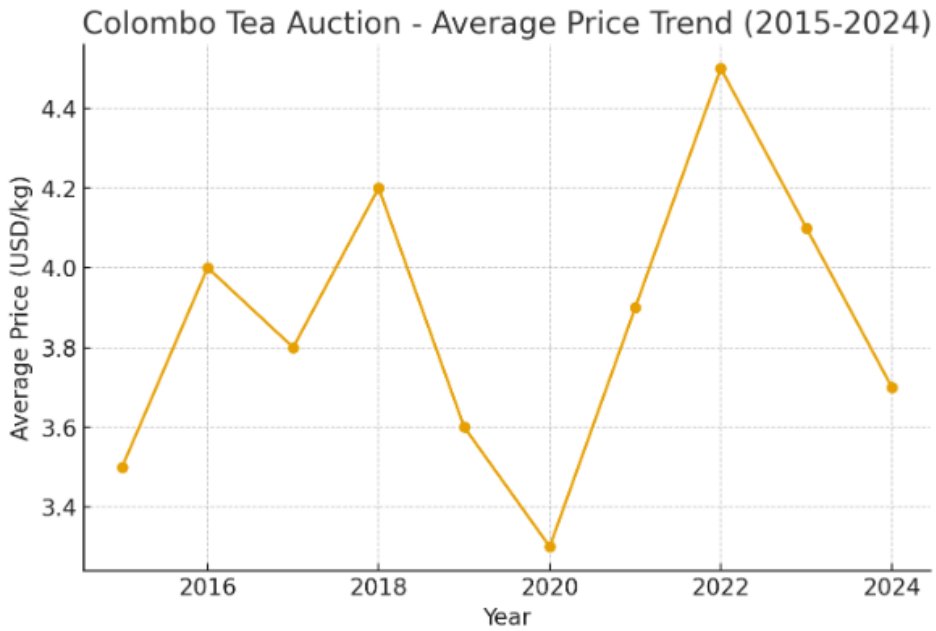

Figure 1.1 Colombo Tea Auction – Average Price Trend (2015 – 2024)

Tea Exporters Association - History of Ceylon Tea

Cost shocks in the recent past merely augment planning issues: in May 2024, producers warned that a 70% wage increase would raise estate spending by ~45%, heightening competitiveness pressures versus Kenya and India yet again making timely analytics worth their weight in gold for pricing and strategy.

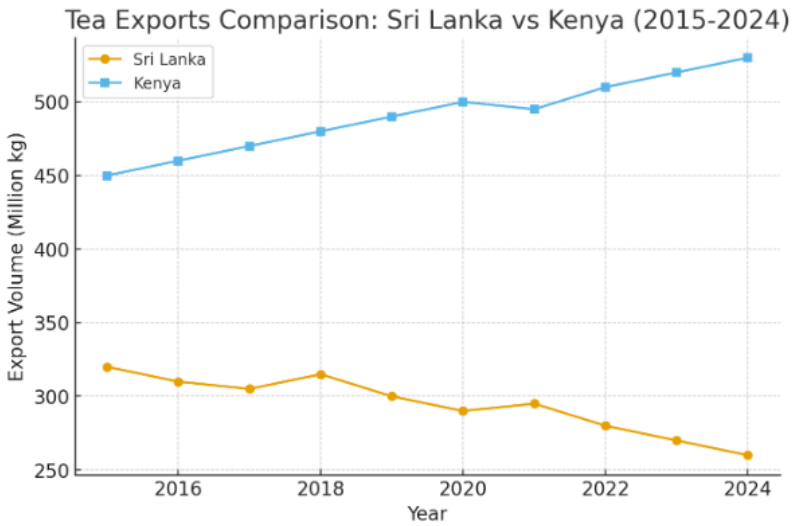

Figure 1.2 Tea Exports Comparison: Sri Lanka vs Kenya (2015 – 2024)

Within this context, peer-reviewed and gray literature regarding Sri Lankan tea price modeling and the like sources suggests different methodological strands

1.1.1 Time-series & Regression baselines.

Hettiarachchi & Banneheka compared traditional time-series regression with Artificial Neural Networks (ANNs) against Colombo auction prices and found ANNs captured nonlinear behavior which linear methods didn't it was the first sign that machine-learning can handle the difficult, sometimes chaotic tea price dynamics. Their article ("Time series regression and artificial neural network approaches for forecasting unit price of tea at Colombo auction") is highly cited in Sri Lankan forecasting literature.

Induruwage, Tilakaratne, & Rajapaksha ("Forecasting Black Tea Auction Prices by Capturing Common Seasonal Patterns") identified common seasonal unit roots and seasonal co integration among Colombo, Kolkata, and Guwahati monthly auction series, making the Seasonal Error Correction Model (SECM) as well as companion Vector Error Correction (VEC) specifications feasible. Their finding confirms that season cycles are the dominant forces and that multi-market linkages matter for Sri Lankan prices.

1.1.2 ARIMA/SARIMA and univariate forecasting.

Some Sri Lankan authors use Box-Jenkins methodology to project auction prices and production. Hameem et al. (and parallel Sri Lankan studies) illustrate that SARIMA models are efficient at identifying elevation-related seasonality (high/mid/low-grown), but weak at pinpointing the impact of exogenous shocks (policy regime changes, input shocks, or FX volatility) i.e., at times of a multi causal data-generating process. Parallel trend/forecasting studies on Sri Lankan tea production also base themselves on uni variate ARIMA/SARIMA, supporting the requirement for isolating drivers other than lagged price/volume.

1.1.3 Multivariate and data-mining methods.

Accepting the limitations of univariate models, B.P. Wickramasooriya's MSc thesis ("Sri Lanka Tea Auction Price Forecast by Using Data Mining Techniques," University of Moratuwa) added weather variables, export quantity, exchange rate, and crude oil price to regression-based data-mining and obtained ~85% prediction accuracy evidence that external drivers improve projections dramatically.

1.1.4 High-level ML/AI and hybrid models.

More recent Sri Lankan export-forecasting work has compared SARIMA, Gaussian Hidden Markov Models, and MLPs, where MLPs perform well for complex, nonlinear patterns in export levels and price series; this is consistent with broader agricultural forecasting literature where regime changes and nonlinearities are more accurately described by hybrid ARIMA-ANN or single-deep-learning models than with linear baselines.

1.1.5 Explainable AI (XAI) for agritea applications.

Mampitiya, Sumanasekara & Rathnayake (2025) created the first explainable model for Sri Lankan (Ceylon) tea yield using CatBoost with SHAP explanations of meteorological, soil, and fertilizer inputs. Price relative to yield-focused however, their Smart Agricultural Technology paper demonstrates (i) the capability to integrate heterogeneous agronomic information, and (ii) how explain ability of the model can identify direction and magnitude of driver’s abilities that can be directly transferred to price and demand modeling where stakeholder trust hinges on explanations.

Along with forecasting mechanics, a number of research articles and reports identify exogenous forces that need to be engineered into Sri Lankan price models:

Global demand & oversupply cycles. Tea & Coffee Trade Journal's Global Tea Report (2024) underscores that oversupply in mainstream teas continues to erode producer revenue despite continued consumer demand for premium segments suggesting structural headwinds that purely domestic models will miss unless they leave off global inventories and rival origins.

Macro-commodity linkages. Commodity Markets Outlook (2024/2025) of the World Bank and supporting data show that broad commodity cycles (energy, fertilizers) and global growth decelerations can compress producer margins and affect auction prices through cost and income channels; such exogenous indices add realism to projections and scenario analysis.

Climate volatility. Decades of Sri Lankan and foreign research (e.g., Wijeratne 1996; follow-up overviews) link tea productivity and quality to heat stress, rainfall aberrations, and the frequency of drought. An MDPI Agronomy review (2021) and related assessments imply that climate change generates both shifts in climatic suitability and variability in quality/biochemical profiles; these effects filter through to prices via supply shocks and quality premiums/discounts, and thus climate covariates and lagged agro-weather indicators are required in forecasting systems.

Supply dominated by smallholders. Nine out of 13 million tea workers worldwide are smallholders, accounting for ~60% of global tea (IISD Global Market Report: Tea, 2024). Whatever decision-support tool is created to stabilize income and enhance bargaining positions needs to be designed for accessibility and fairness, not just for major exporters.

Combining, the literature shows a unified picture: (i) Sri Lankan tea prices are seasonal and move together with auctions locally; (ii) uni variate models will perform reasonably well in peaceful regimes but collapse under shocks exogenous; (iii) multivariate ML and XAI will include weather, macro, and market covariates and provide explainable signals; and (iv) yet, no industry-standard, real-time, AI-powered platform exists that aggregates these models into an open, stakeholder-friendly dashboard to facilitate strategy and risk management.

1.2 Research Gap

Despite global evolution of digital platforms in agricultural and commodity markets, the Sri Lankan tea industry is still largely in a relatively traditional framework with significant reliance on Colombo Tea Auction as the central system for price discovery and transactions. Even though the auction system is globally renowned and has existed as a structured trade platform for decades, it is susceptible to several limitations. It is primarily centralized and largely out of sight for mid-scale producers and smallholders, which represent the overwhelming majority of the supply chain but lack direct access to timely market data. Previous studies in agricultural economics and research on commodity digitization have shown that how farmers and producers in other countries benefit significantly from smart dashboards and real-time analytics platforms providing forecasting, demand trends, and open pricing. Notwithstanding its international relevance, hardly any attempt has been made to apply such tools for the Sri Lankan tea sector.

Apart from that, studies on precision agriculture, big data, and digitalization of commodity markets (e.g., India, Kenya, and China studies) demonstrate that integrating AI-driven dashboards with predictive analytics enables better decision-making from cultivation planning to export strategies for all concerned. In contrast, the Sri Lankan tea system still lags behind a comprehensive digital platform to serve as an intermediary among farmers, brokers, exporters, and policymakers. Existing systems either are fragmented (with exporters relying on their own internal forecasting models) or are single-player-based (excluding small and medium players). This creates an information asymmetry, where large-scale market players possess privileged information and smallholders are disadvantaged, relying on auction outcomes without expectation.

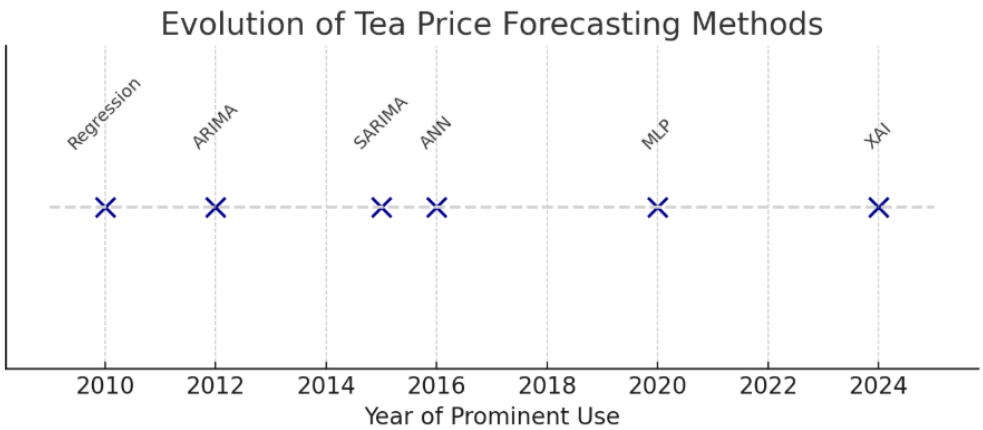

Figure 1.3 Evolution of Tea Price Forecasting Methods

The research gap, therefore, is the absence of an integrated, stakeholder-accessible smart dashboard that pools market information, applies forecasting models, and provides real-time, actionable information to all levels of the industry. No platform

currently available in Sri Lanka offers a web-based decision-support system, geared towards the tea market, which simultaneously addresses transparency, accessibility, and inclusivity. This leaves a glaring void where tech innovation can democratize market intelligence and enable the entire value chain from small tea producers to large exporters to have an equal opportunity for information and sustainable development of the sector.

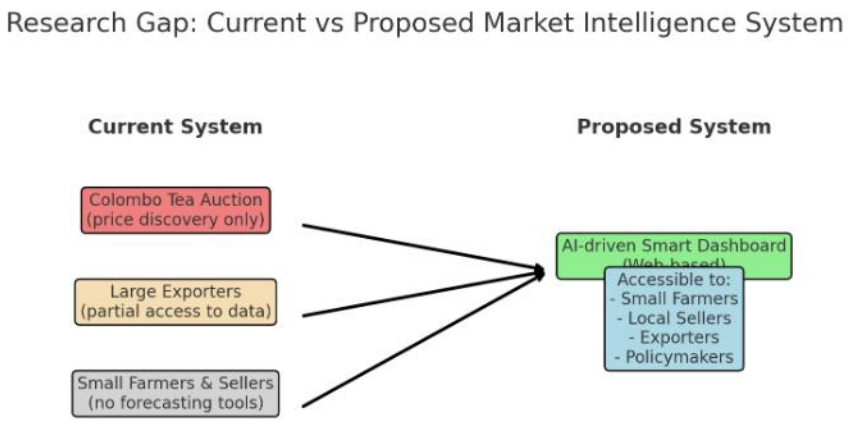

Figure 1.4 Current vs Proposed Market Intelligence System

1.3 Research Problem

Sri Lankan tea industry, even though globally renowned as the world's largest producer of orthodox tea, still relies on the Colombo Tea Auction as its primary price determination and trading mechanism. Even though the auction system has been globally hailed as resilient and formalized in its process, it is centrally placed, traditional, and secretive in its nature, imposing limitations on the total tea value chain. Small-scale farmers, who account for over 70% of country production, remain widely dependent on brokers and intermediaries for information, with no access to real-time market data, demand forecasts, and clear-cut pricing indications. This puts them in a position where they are making decisions based on outdated or partial information, placing them at a structural disadvantage compared to large-sized exporters who use private forecasting devices.

Even while other top agriculturally and commodities-producing countries such as India, Kenya, and China have begun adopting AI-enabled forecasting platforms, digital dashboards, and precision market intelligence platforms, Sri Lanka is behind by leaps in adopting such digital transformations to its tea value chain. Its existing system is exclusionary in design exporters operate within isolated digital silos, while smallholders and mid-scale farmers depend almost entirely on post-auction results. This

imbalance helps facilitate information asymmetry in suppressing fair growth, fair competition, and long-term industry sustainability.

The problem is also exacerbated by increasingly volatile international prices of tea, increasing international demand uncertainty, and pressure from other competing tea-producing nations that already enjoy modernized commodity management systems. In the absence of an integrated, accessible, and inclusive platform, Sri Lanka will most likely experience reduced competitiveness in international markets, persistent marginalization of small-scale farmers, and missed opportunities in leveraging big data and AI for optimizing local and global market strategy.

Hence, the focal research problem is the absence of an integrated AI-driven smart dashboard specific to Sri Lankan tea and capable of democratizing forecasting, demand pattern, and transparent price insights' access for the entire supply chain and for market analysis. Its solution is critical not only to enrich mid-scale producers and smallholders but also to elevate Ceylon Tea's global brand position with innovation, inclusivity, and sustainable market knowledge.

2. Objectives

2.1 Main Objective

To deliver an integrated real-time AI-driven decision support system with an interactive smart dashboard that incorporates multi-source market data, applies state-of-the-art forecasting models, and enables intricate strategy development for the Sri Lankan tea industry. The system must deliver accurate sales market analysis and price forecasting for international as well as local tea markets, while supporting price risk management along the value chain to improve transparency, inclusivity, and long-term competitiveness.

2.2 Specific Objectives

1. To assess the current price discovery and sales arrangements in Sri Lanka's tea industry

Review the role of the Colombo Tea Auction and identify areas of deficiency in price risk management and access to sales intelligence.

2. To explore global best practice in digital commodity platforms and AI forecasting

Investigate the use of dashboards and forecasting applications in India, Kenya, and China and identify transferable characteristics for Sri Lanka.

3. To collect and integrate multi-source market intelligence for sales forecasting and risk management

Leverage internal (auction results, production, records of exporters) and external (weather, global indices, demand of consumers, competitor data) sources.

4. To build and implement AI-based predictive models

Develop forecasting models for demand trends and tea prices using time series and machine learning algorithms.

Integrate risk analysis tools such as volatility monitoring, Value-at-Risk (VaR), and scenario testing.

5. In order to create an interactive and inclusive smart dashboard

Provide real-time data, price risk alerts, and sales market analysis in an intuitive interface.

Provide role-based access by smallholders, brokers, exporters, and policymakers.

6. To evaluate performance of the proposed system

Conduct pilot testing among principal stakeholders.

Evaluate increases in price risk management, sales decisions, and overall market intelligence inclusiveness.

3. Methodolgy

The research methodology is systematic and cyclical, beginning from the problem identification and requirement analysis where the key problems in Sri Lankan tea industry such as price volatility, lack of access to real-time data, and inefficiency in forecasting are clearly identified. The stakeholder needs, encompassing exporters, indigenous traders, government representatives, and tea boards, are established to identify the system needs of the proposed smart dashboard, emphasizing the features of data integration, forecasting precision, user experience, and risk insights. The next phase consists of data collection, integrating primary data (through market research, stakeholder interviews, and expert view) and secondary data, internal sources such as the Sri Lanka Tea Board, Colombo Tea Auction price levels, and estate production reports, and external sources such as international tea market indices, international commodity exchanges, climatic conditions, and policy or regulatory announcements. Both structured data (such as price data, export volume) and unstructured data (such as news articles, global reports) are gathered in order to include an entire dataset.

The gathered data is pre-processed through cleaning (removal of missing data, duplicates, and noise), transformation (conversion of currency, normalization, and time synchronization), and feature engineering (consideration for climate effect, seasonal demands, and competitor trends) before a combined data warehouse is produced to serve as the system's input layer. The next step is model building, beginning with descriptive analysis of historical price behavior and volatility and then predictive

analysis using regression models (Linear, Ridge, Lasso) and time series models (ARIMA, SARIMA, Prophet, LSTM) to enhance the accuracy of predictions. At the same time, risk analysis models such as Value at Risk (VaR), Monte Carlo simulations, and sensitivity analysis are built to quantify risk levels. These models are trained, tested, and validated by cross-validation processes.

The next phase of dashboard development and designing involves integrating trained models in a scalable backend platform (Node.js/Python) with the implementation of an interactive and user-friendly frontend (React.js along with data visualization libraries like D3.js and Plotly). The key features of the dashboard involve real-time data updates, market trend visualization, price forecasting modules, and risk alerts accompanied by actionable suggestions to make it accessible to various stakeholder groups. In the system integration and validation phase, APIs are interfaced for real-time feeds (auction prices, global indices, weather), and the system is tested intensively for data load performance, predictability, and usability based on stakeholder feedback.

The system is then implemented in a secure cloud environment (AWS, GCP, or Azure) with scalability for growing data and user load, with compatibility on mobile and desktop platforms. The assessment and impact evaluation stage tests forecasting accuracy based on MAE, RMSE, and MAPE metrics and the efficacy of risk management in curbing price volatility. Usability, decision-making improvement, and overall economic and policy-level impact on the tea industry are tested through stakeholder feedback. Finally, the project has an improvement loop in place with continuous model updating of forecasting models based on new data, optimization of features utilizing user feedback, and addition of advanced AI-based recommendations and scenario-based risk simulations to ensure long-term sustainability and timeliness of the system.

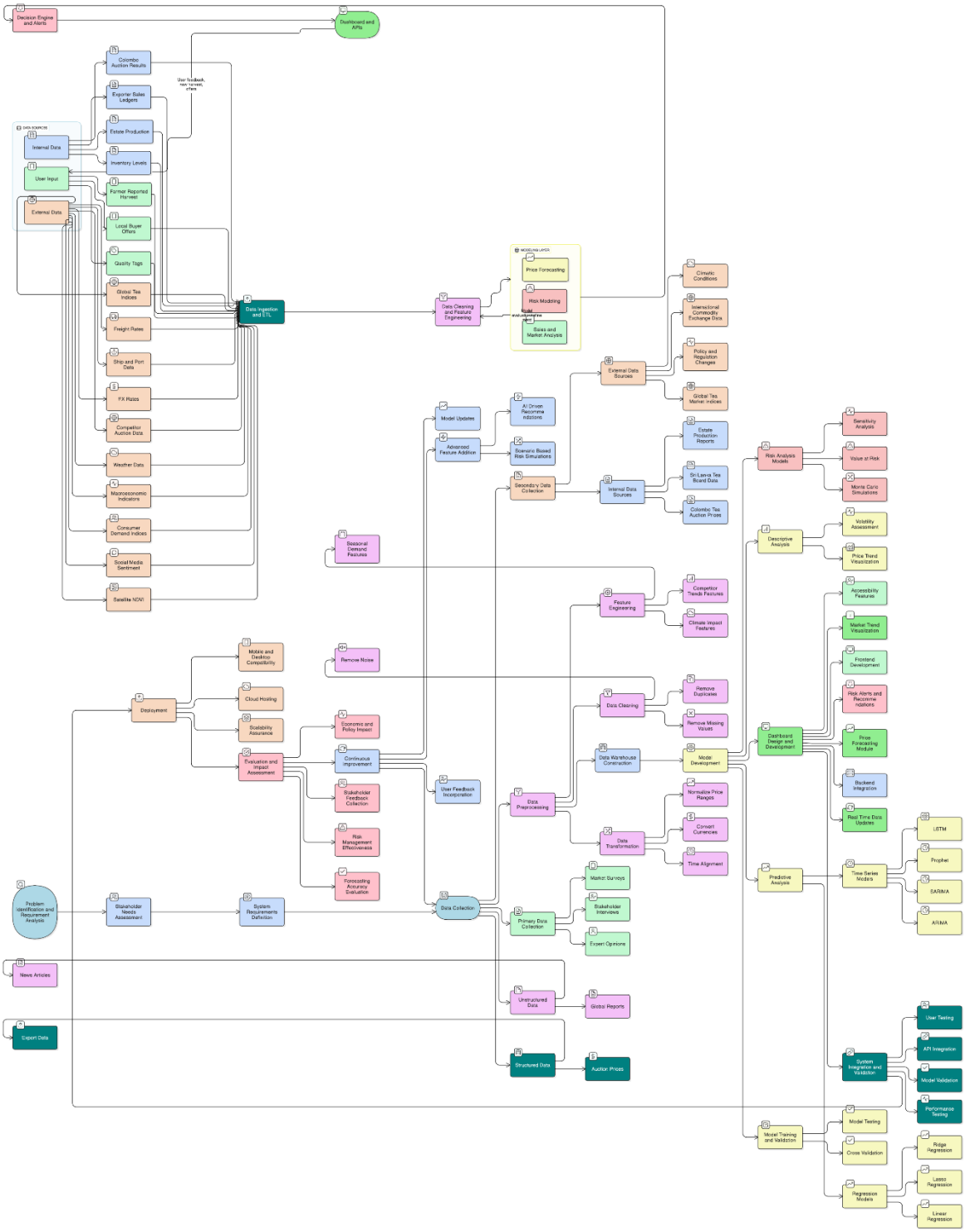

Figure 3.1 System Diagram for Overall System

3.1 Software Solution

The proposed software solution looks to address the issue of the Sri Lankan tea industry by creating an AI-based smart dashboard with real-time price predictions, risk assessment, and decision-making capabilities. The methodology is systematic with development process phases as feasibility phases, requirement collection and analysis, user data preparation, and implementation, all under the directions of the Agile development methodology.

3.1.1 Development Process

The system is constructed upon the Agile software development process, which emphasizes iterative improvement, continuous stakeholder engagement, and incremental provision of working pieces. Compared to the traditional method, where developers must complete each stage in sequence and have no opportunity to revisit earlier stages when requirements are revised, Agile leaves space for elasticity and responsiveness. Agile is focused on being responsive to change, so requirements can be modified during development and changes can be incorporated without having to do the entire process over again. This ensures the system is still aligned to stakeholder requirements as new information arises.

Development is divided into sprints with each sprint covering particular deliverables such as data integration, training forecasting models, risk analysis tools, or dashboard visualization. Prototypes are presented to stakeholders (exporters, tea board members, traders) at the end of each sprint to receive feedback and improve. Agile practices not only reduce risks of project misalignment but also facilitate ongoing improvement, which guarantees that the final solution offers both functional accuracy and stakeholder satisfaction.

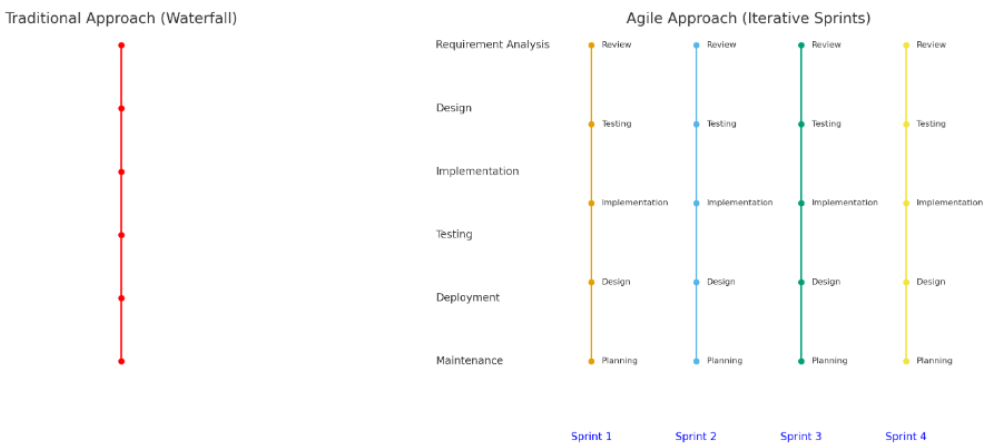

Figure 3.2 Agile vs Waterfall Approaches

3.1.2 Feasibility Study

Feasibility study assesses the feasibility of the proposed solution:

Technical Feasibility: Libraries such as React.js for frontend, Node.js/Python for backend, and machine learning libraries (TensorFlow, Prophet, Scikit-learn) lend technical merit. Cloud platforms (AWS, GCP, Azure) facilitate real-time scalability.

Operational Feasibility: Stakeholders are continuously in need of real-time market knowledge. Agile's nature of iterative development ensures they are consulted regularly, so the system is ready for adoption.

Economic Viability: Despite heavy upfront investment, benefits such as improved risk management, reduction of losses via price volatility, and export competitiveness ensure long-term fiscal viability.

Legal/Ethical Viability: Utilization of data meets ethical requirements and regulatory guidelines and respects stakeholder anonymity and responsible deployment of AI.

3.1.3 Requirement Gathering and Analysis

Requirements in Agile are addressed as user stories and use cases, collected from diverse stakeholders through surveys, interviews, and workshops.

Functional Requirements: Real-time data integration, forecasting feature, trend analysis visualization, and risk alerting. The dashboard must be able to support interactive charts, compare functionalities, and recommendation modules.

Non-Functional Requirements: Accessibility, scalability, usability, data security, and performance dependability with heavy data loads.

Stakeholder Analysis: Exporters need price forecasting for coordinating shipping schedules, traders need real-time updates for purchase decisions, and regulators need risk and policy-level analysis. These demands are continually aligned through development cycles.

3.1.4 Dataset

The dataset is the epicenter of the predictive functionality of the dashboard.

Primary Data: Collected through industry surveys, stakeholder interviews, and risk factor expert input.

Secondary Data: Internal sources: Sri Lanka Tea Board statistics, Colombo Tea Auction prices, production reports at the estate level.

External sources: International commodity indices, competitor nations' production levels, international auction reports, weather/climate data, and government policies.

Structured data (price series, export volumes, production levels) and unstructured data (news, reports, and regulatory notices) are both combined. Pre-processing operations cleaning, normalization, feature engineering, and transformation are performed prior to feeding the consolidated dataset into the model pipeline.

3.1.5 Implementation

Implementation phase follows Agile sprints in which a functional portion of the system is developed every sprint. The backend is developed with Python/Node.js to interlink forecasting and risk analysis models, and the frontend is developed with React.js with visualization libraries (D3.js, Plotly) for interactive user interface. Machine learning and time series models (Linear Regression, Ridge, Lasso, ARIMA, SARIMA, Prophet, LSTM) are defined and tested in iterations. Risk estimation frameworks such as Value at Risk (VaR) and Monte Carlo simulations are incorporated to quantify uncertainties. The solution is deployed on a secure cloud platform to ensure maximum scalability and real-time updates, with APIs linked to real-time data sources such as tea auctions, commodity markets, and climate services. Testing unit, integration, system, and user acceptance is conducted at the end of each sprint, delivering incremental functionality and reliability.

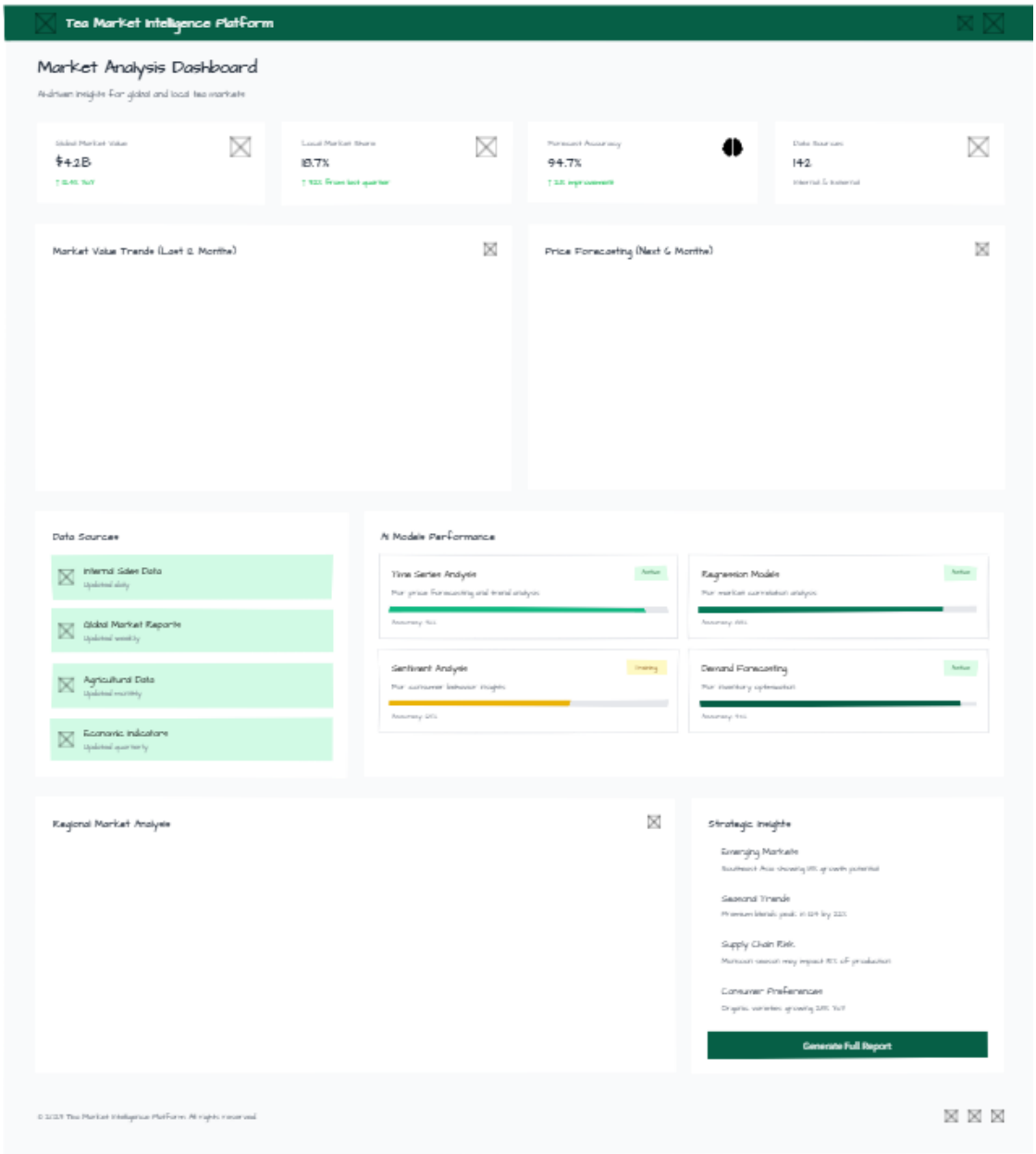

Figure 3.3 Wireframe for the Proposed Smart Dashboard

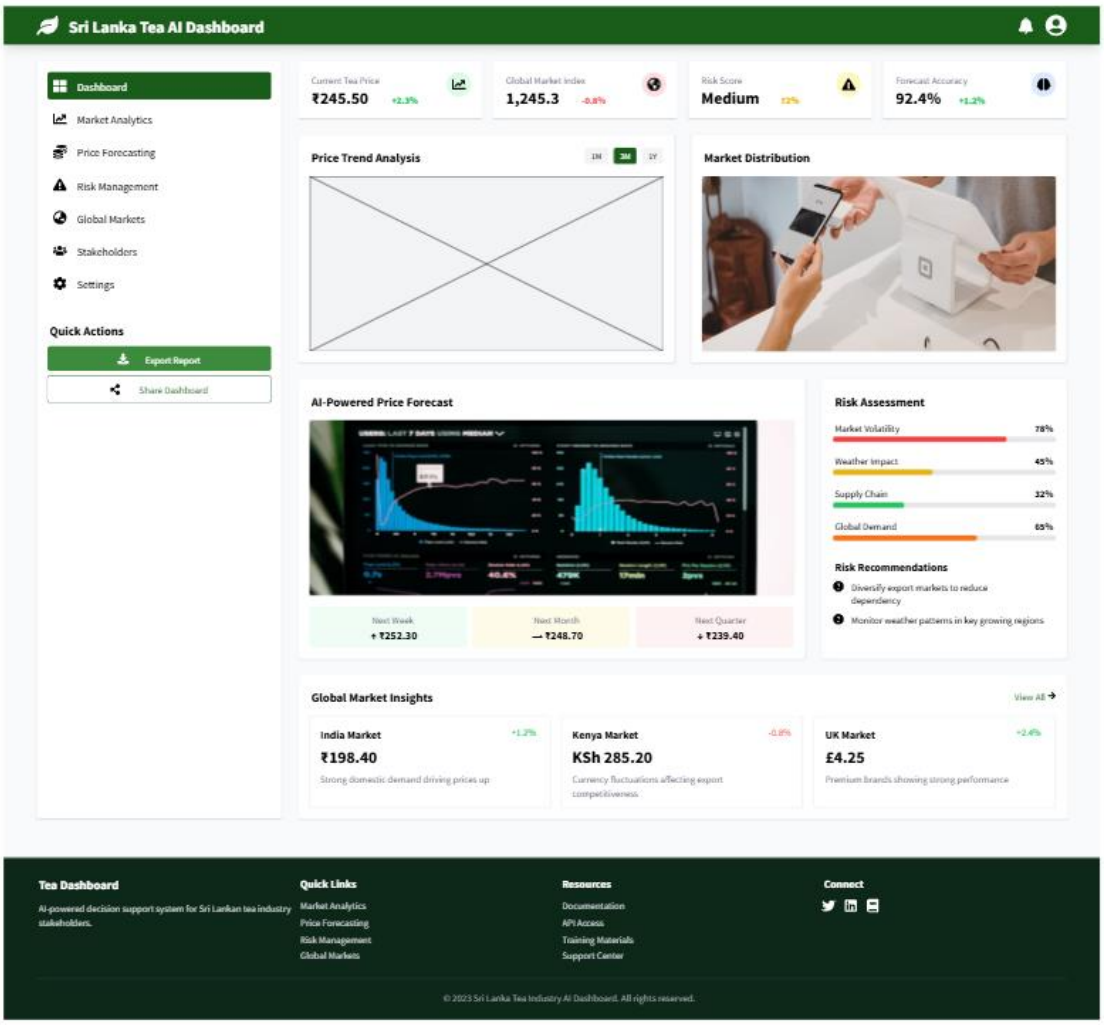

Figure 3.4 UI for the Proposed Smart Dashboard

3.1.6 Testing

To ensure the reliability, accuracy, and usability of the smart dashboard, the system undergoes four levels of testing throughout the Agile development cycle:

Unit Testing

Describes the process of testing individual units or modules of the system, such as one machine learning algorithm, API endpoint, or visualization component.

Ensures that each function is performing as anticipated individually before integrating them.

Integration Testing

Ensures different modules of the system (e.g., data ingestion, preprocessing, model prediction, and visualization dashboard) play in harmony with one another.

Detects defects in the communication between parts for smooth data flow across the pipeline.

System Testing

Executes on the integrated system for checking whether the whole dashboard meets both functional and non-functional requirements.

Checks performance, security, scalability, and reliability at different data loads and usage cases.

User Acceptance Testing (UAT)

Involves stakeholders such as regulators, traders, and exporters testing the dashboard in real working settings.

Verifies that the solution is sufficiently practical, user-friendly, and provides insightful decision-making data before deployment.

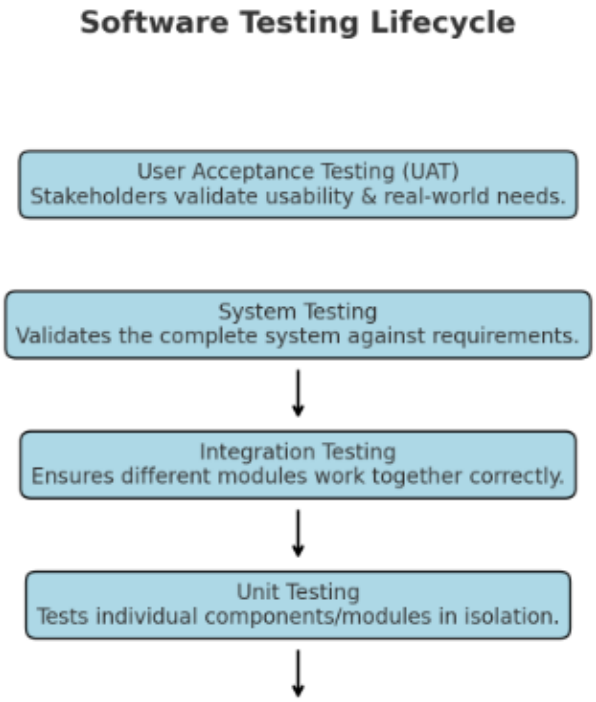

Figure 3.5 Software Testing Lifecycle

4. Project Requirements

The proposed AI-powered smart dashboard for Sri Lanka's tea industry requires a blend of user, software, functional, and non-functional requirements that will ensure accuracy, usability, and flexibility. Since the ultimate goal is enhancing market forecasting and risk management, market analysis also forms an important requirement dimension.

4.1 User Requirements

The system should provide a convenient dashboard with simple-to-use design and easy navigation.

Stakeholders (exporters, traders, estate managers, regulators) should be able to see market trends and price projections in real-time.

There needs to be the functionality of comparing domestic prices against international indices so that decision-making can be informed.

Risk notifications and alerts must be provided by the system when there is abnormal price volatility or weather risk.

Reports and visualization need to be exportable (PDF, Excel) for trade or policy purposes.

Multi-device availability (desktop, tablet, mobile) with role-based access control.

4.2 Software Requirements

Frontend: React.js with D3.js/Plotly for interactive visualizations.

Backend: Python/Node.js for data processing and ML model integration.

Databases: SQL/NoSQL storage and centralized data warehouse for merged datasets.

Machine Learning Libraries: TensorFlow, Prophet, Scikit-learn, and LSTM for prediction.

Data Sources & APIs: Colombo Tea Auction data, Sri Lanka Tea Board reports, global commodity markets, competitor country indices, and climate APIs.

Cloud Infrastructure: AWS/GCP/Azure for secure hosting, scalability, and availability.

Market Data Integration Tools: Web scraping, REST APIs, and ETL pipelines for structured and unstructured data extraction.

4.3 Functional Requirements

Market Analysis Capabilities:

Provide descriptive analytics (historical price trends, volatility patterns).

Perform predictive analytics (regression, ARIMA, Prophet, LSTM models for price forecasting).

Provide comparative analysis (local vs. global price movements, competition performance).

Include sentiment and news analysis from unstructured sources to include global market drivers.

Core Functionalities:

Incorporate real-time data integration from diverse sources.

Interactive visualizations: heatmaps, line charts, dashboards.

Quantifying risk tools (Value at Risk, Monte Carlo simulations).

Custom export and reporting functionality.

Risk notifications/alerts on unusual variations.

Model auto-update as new data arrives.

4.4 Non-Functional Requirements

Usability: Plain design with minimal training required.

Performance: Large volume datasets with low latency (<5 seconds for queries).

Scalability: Support increase in user base and increasing datasets as more markets are added.

Security: End-to-end encryption, user authentication, and role-based authorization.

Reliability: High uptime (99.9%) with disaster recovery mechanisms.

Portability: Use on devices (desktop, tablet, mobile).

Maintainability: Modular architecture to enable easy updates and upgrades.

Interoperability: Interoperability with other government, financial, or commodity platforms.

Compliance: Adherence to international data handling and ethical AI guidelines.

5. Gantt Chart

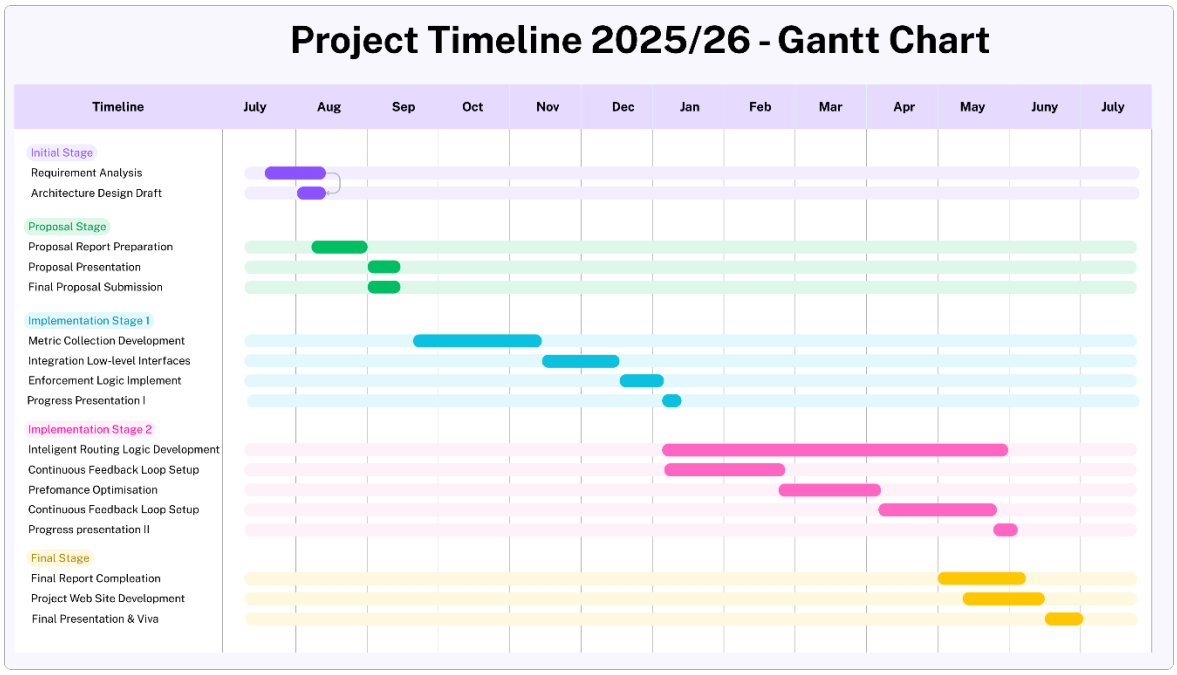

Figure 5.1 Gantt Chart

6. Work Breakdown Diagram

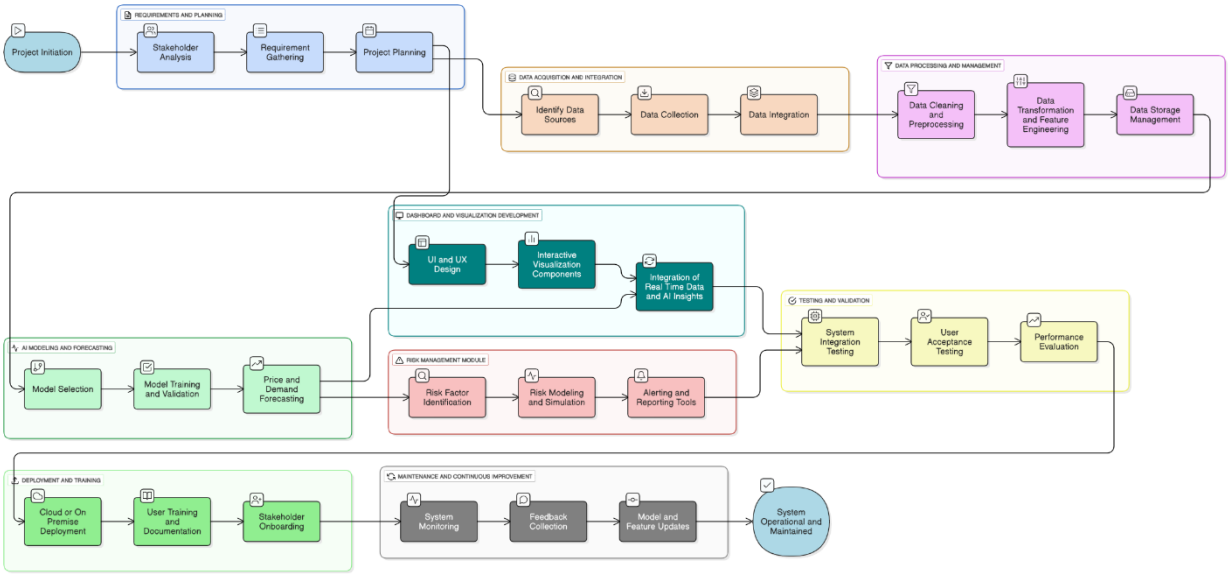

Figure 6.1 Work Breakdown Diagram

7. Description of Personal and Facilities

Personnel List

Researcher (Student) - Responsible for research, data collection, system design, programming, and report preparation.

Supervisor and Co-Supervisor - Maintain research standards and offer academic guidance.

Data Analyst - Assist in cleaning, processing, and verifying tea market datasets.

Dilmah Factory Management - Offer access to production and market data and facilitate industry insights.

Industry Consultant - Share expert opinions on tea market trends, commercialization, and best practices.

Support Staff – Join on field visits, administrative tasks, and logistics arrangements.

Facilities:

University Facilities

The university is equipped with computer labs, research libraries, internet, and academic databases. These facilities support coding, testing, literature review, and regular discussions with supervisors.

Dilmah Kahawatta Tea Factory

The factory, as the industry partner, provides primary and secondary data on tea production, pricing, and sales. Discussions with factory officers and specialists ensure the dashboard's alignment with real-world industry practices.

Software and Tools

The project leverages Python for analysis, the MERN stack (MongoDB, Express, React, Node.js) for dashboard development, and visualization libraries such as Plotly or D3.js. Cloud platforms can also be used for deployment and real-time analysis.

8. Budget & Budget Justification

Component	Cost
Travelling	LKR 18,000
Web Application Hosting	LKR 9,000
Domain Registration	LKR 5,000
Software Tools & Licenses	LKR 6,000
Data Collection & Cleaning Tools	LKR 4,000
Miscellaneous Costs	LKR 3,000

Table 8.1 Budget

Total Estimated Budget: LKR **45,000**

The total amount budgeted for execution of this project is LKR 45,000, divided wisely to be done within cost effectiveness without sacrificing on all the major requirements of the AI-based tea price prediction and market insights dashboard. Travelling expenses consume half of the amount, LKR 18,000, because two days field visits compulsory from Colombo to Kahawatta plantation are required. These site visits will enable us to collect primary data, validate tea price information, and deal directly with plantations'

stakeholders to ensure the dashboard is grounded in real-world facts as well as global market statistics. For the cyber infrastructure, there is LKR 9,000 for six-month web hosting that covers a mid-level secure cloud hosting package (roughly LKR 1,500/month) to ensure testing, deployment, and fluent access to the platform. LKR 5,000 is also provided for domain name registration (.com or .lk), which not only provides professionalism but ensures easy access to the system for exporters, brokers, and researchers as well.

For enhancing system performance, LKR 6,000 has been allocated for software licenses and tools, including premium APIs for commodity and forex market data and limited AI/ML service subscriptions for facilitating sophisticated regression and time-series forecasting models. Further, LKR 4,000 has been allocated for data collection and cleansing tools, including the costs of structured storage, processing, and purification of local and global tea market datasets. Finally, LKR 3,000 is reserved for miscellaneous items such as printing, binding, and presenting the final report, meeting the small, unexpected project needs. In total, the budget of LKR 45,000 provides a fairly balanced and reasonable allocation that will allow the project to be effectively implemented without any compromise on data quality, system reliability, or professional presentation.

9. Commercialization

9.1 Target Audience

The AI-powered tea market analysis system is designed to serve the interests of a wide range of stakeholders in the Sri Lankan tea industry by ensuring equal access to predictive data and strategic recommendations. The primary target audience is:

Smallholder Tea Farmers

They constitute the majority of tea growers in Sri Lanka. They are likely to forego formal market data, and thus are highly susceptible to price volatility. The platform provides them with real-time price forecasts, demand trends, and strategic analysis to optimize harvest and selling time for enhancing revenues' stability.

Local Tea Vendors and Cooperatives

Regional and local suppliers are benefits to companies by way of market demand, seasonality, and good procurement information. This facilitates wise pricing and stock control decisions that maximize profitability.

Large Exporters and Tea Firms

Large tea exporters can leverage predictive analytics to plan export volumes, identify profitable markets, and avoid exposure risks to global price fluctuations and policy adjustments. Scenario analysis is also provided on the platform for examining the impact of macroeconomic and climatic influences on pricing strategies.

Policymakers and Industry Regulators

Government agencies, agriculture commissions, and policymakers can utilize summated data to establish support programs, stabilize the markets, and foster sustainable development in the tea sector.

9.2 Market Space

The Sri Lankan tea industry is a deeply rooted global commodity, with the name of Ceylon Tea being globally recognized as superior quality. While global in scope, the industry faces price fluctuations, competition from competing tea-producing countries, and broken access to market intelligence. The multi-faceted AI-based system occupies a niche market position in the agritech and decision support systems market, bridging the gap between traditional tea trading conventions and evidence-based market intelligence.

Domestic Market

The platform engages an under-served segment of smallholders and local traders who lack access to formal, actionable information. Democratizing access to market intelligence, it increases local business performance, reduces revenue risk, and promotes sound decision-making.

Global Market

Exporters who are willing to enhance competitiveness can use predictive analytics to make strategic decisions, aligning the timing of auctions, blending, and reaching overseas buyers. The system's capability to integrate international price indices and macroeconomics makes it a strategic tool for maximizing international markets.

By targeting such stakeholder groups, the system creates value along the value chain of tea, facilitating sustainable development, market clarity, and strategic innovation within one of Sri Lanka's most significant export sectors.

Conclusion

The proposed AI-driven tea market analysis system represents a significant advancement in the digitization and modernization of the Sri Lankan tea industry. By integrating diverse internal and external datasets ranging from Colombo Tea Auction micro data, exporter shipment logs, and domestic market indicators to global price indices, macroeconomic variables, and climatic factors the system offers a holistic view of the market. Traditional methods, such as reliance on historical patterns or fragmented manual analysis, are limited in scope, particularly for smallholder farmers and local vendors who lack access to structured market intelligence.

Through the application of advanced forecasting models, including SARIMA, MLPs, and gradient-boosted decision trees, the system provides robust, real-time predictions of tea prices, accounting for both seasonal patterns and external market shocks. Explainable AI components, including SHAP values and scenario simulation tools, enhance transparency and interpretability, enabling stakeholders to understand the key drivers behind price movements and make informed decisions.

The interactive smart dashboard consolidates these predictive insights into a user-friendly interface accessible to all stakeholders, from smallholders and local traders to large exporters and policymakers. This democratization of market intelligence addresses the persistent issue of information asymmetry, enabling equitable access to actionable insights. Consequently, stakeholders can optimize harvest timing, pricing strategies, auction participation, and export planning while minimizing risk exposure due to market volatility, climatic uncertainties, and global demand fluctuations.

By empowering the entire value chain with data-driven decision support, the system enhances the competitiveness of Sri Lanka’s tea industry on the global stage. It also promotes resilience, sustainability, and efficiency, aligning with broader objectives of economic growth and technological adoption in agribusiness. Overall, this research demonstrates that AI-powered decision support can bridge long-standing information gaps, improve revenue stability, and contribute to a strategic, evidence-based approach to market engagement.

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